Does the Visual World Paradigm show evidence of referential adaptation? Comparing referential adaptation across two eye-tracking experiments.

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Ambiguous pronoun resolution is guided by multiple constraints, including a well-established subject bias: listeners preferentially interpret pronouns as referring to the grammatical subject of the preceding clause (Arnold et al., 2000; Jarvikivi et al. 2005; Kaiser, 2009). Recent behavioral evidence demonstrates that comprehenders can modulate this bias when exposed to pronoun-referent patterns that violate their expectations (Johnson & Arnold, 2023; Ye & Arnold, 2023; Roy et al., 2025). This phenomenon is called referential adaptation. However, it remains unclear whether such effects of adaptation can be seen during online processing or whether adaptation only influences offline judgments and final interpretations. We report two visual world eye-tracking experiments that examine online evidence for referential adaptation, revealing how trial structure affects the detectability of adaptation effects.

Method: Both experiments (N=40 each) used a between-subjects exposure manipulation (subject-referent vs. nonsubject-referent pronouns in 28 filler items) crossed with a within-subjects target manipulation (subject vs. nonsubject targets in 16 critical ambiguous items). In critical trials, participants heard stories about two same-gender characters (e.g., "Liz went shopping with Ana at the new mall. She bought a new watch because her old one was broken.") while viewing images where only one character matched the disambiguating information. We analyzed empirical logit-transformed fixations to the subject character during the ambiguous period (300ms from the pronoun onset to 300ms post-disambiguation).

Example Stimuli, **Subject exposure story:** Will won a trophy with Liz at the spelling bee competition. He was very proud for spelling all the words correctly(Fig. 3). **Nonsubject exposure story**: Will won a trophy with Liz at the spelling bee competition. She was very proud for spelling all the words correctly(Fig. 4). **Critical ambiguous story:** Liz went shopping with Ana at the new mall. She bought a new watch because her old one was broken.(Fig.5).

Results: Experiment 1 revealed a significant main effect of exposure, $F(1, 17.8) = 5.04$, $p = .038$ in a mixed effects model. Participants in the nonsubject-exposure condition showed reduced subject bias overall compared to those in the subject-exposure condition. The interaction between exposure and target type did not reach significance, $F(1, 14.7) = 2.97$, $p = .106$. Trial presentation in Experiment 1 was inadvertently quasi-blocked: within each list, the first half and second half of trials systematically differed in target type and verb type, creating a block-like structure that may have strengthened cumulative adaptation effects.

Experiment 2 replicated Experiment 1 with improved pseudo-randomization, distributing all types of critical items (balancing target type and verb type) evenly throughout the experiment. Results after running a mixed effects model showed no significant main effect of exposure, $F(1, 20.6) = 0.43$, $p = .517$, and no significant interaction, $F(1, 10.6) = 2.97$, $p = .114$. However, the numerical pattern of the non-significant interaction was identical to Experiment 1 (same F-value), with participants showing greater differentiation between subject and nonsubject targets following subject exposure compared to nonsubject exposure.

Conclusion: The contrasting main effect patterns across experiments—significant in Experiment 1 but absent in Experiment 2—have two possible interpretations. First, the quasi-blocked structure in Experiment 1 may have allowed cumulative adaptation to build over successive trials, making a small main effect detectable that was too weak to emerge under distributed presentation in Experiment 2. Alternatively, the main effect in Experiment 1 may have been artifactual. The identical numerical pattern for the interaction across both experiments ($F = 2.97$), while not reaching significance in either study, suggests potential modulation of adaptation by target type that requires larger samples to detect reliably. These findings provide mixed evidence for online referential adaptation: a main effect emerges under blocked presentation but not under randomized presentation, while highlighting the critical role of trial structure and statistical power in detecting subtle adaptation effects in pronoun resolution.

Figure 1. Exp 1 Results

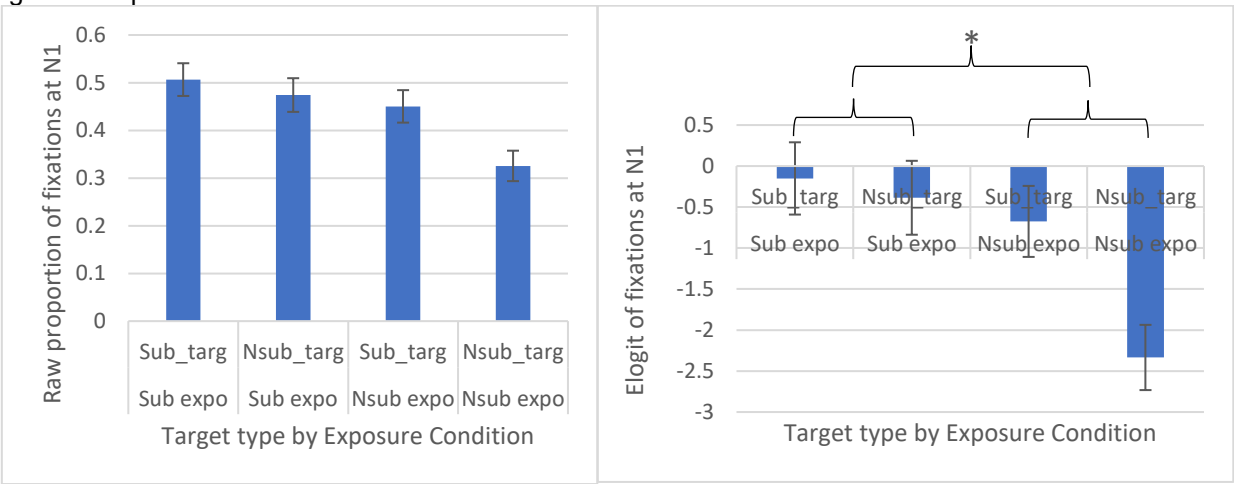
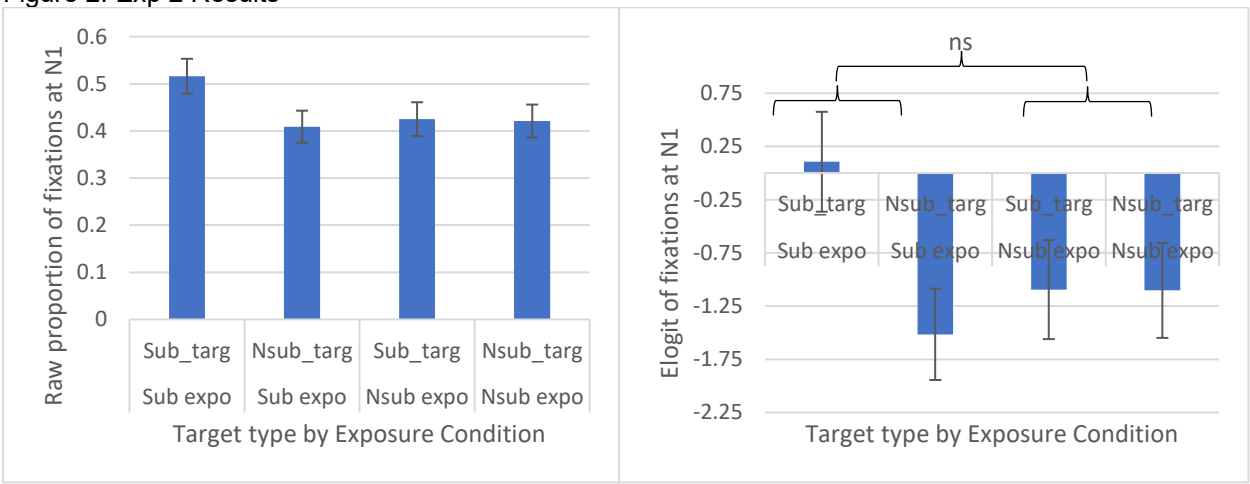


Figure 2. Exp 2 Results



Example stimuli :

Fig.3.Subject exposure Fig. 4.Nonsubject exposure Fig. 5. Critical ambiguous story

References: Arnold, J.E., et al. (2000). The rapid use of…*Cognition, 76(1)* ♦ Jarvikivi, J., et al. (2005). Ambiguous Pronoun Resolution…*Psychological Science, 16(4), 260-264* ♦ Johnson, E., & Arnold, J. A. (2023). The Frequency of referential patterns… *JEP: LMC* ♦ Kaiser, E. (2009). Effects of anaphoric… *DAARC (pp. 121–129). Springer* ♦ Roy, S., et al. (2025). Scrutinizing Referential Adaptation… *Psych Bulletin and Review* ♦ Ye, Y. & Arnold, J.E. (2023). Learning the statistics of.. *Cognition,239.*